Daniel Thomas

516-591-9249 | diggyzar@bu.edu | linkedin.com/in/daniel-g-thomas | github.com/diggygeorge

Education

Boston University Boston, MA

Bachelor of Arts in Mathematics and Computer Science *Expected May 2027*

- **GPA:** 3.91 (Dean's List)
- **Relevant Coursework:** Distributed Systems, Cloud Computing, Databases, Data Structures and Algorithms, Operating Systems, Computer Systems, Linear Algebra

Technical Skills

Languages: Python, Java, Golang, C/C++, TypeScript, JavaScript
Data: SQL (Postgres, MySQL), Snowflake, dbt, MongoDB, Supabase, Prisma
Cloud & DevOps: AWS, Docker, Kubernetes, OpenShift, Linux, Google Cloud Platform, Git
Web & Frameworks: Spring Boot, Flask, Node.js, Next.js, Express.js, React, Tailwind CSS
Certifications: AWS Certified Cloud Practitioner (Passed Sep 2026)

Experience

Software Engineering Intern May 2026 – Aug. 2026

Red Hat - Data and AI *Lowell, MA*

- **Created a company-identification pipeline** using Python and SQL for URL normalization and multi-dataset relational joins to deterministically match domains, achieving a **77% company match rate** to **resolve 110,000+ anonymous monthly web interactions** into 6,000+ companies
- **Aggregated anonymous web traffic** in Snowflake, implementing SQL regex and pattern matching on page types and visit frequency to **categorize raw hits into buying signals** across 40 Red Hat products
- **Containerized an internal Streamlit application** and deployed it on OpenShift/Kubernetes, delivering a self-serve UI that **reduced data querying to under 60s** for cross-disciplinary engineering and marketing teams
- **Engineered an aggregate data product** using dbt on GitLab to **power 3 downstream AI platforms** and scale internal analytics operations

Software Engineer Jan. 2026 – Present

Hack4Impact *Remote*

- **Architected secure REST APIs** using Next.js, Prisma, and Supabase, **implementing role-based authentication** and mutating endpoints to manage event schedules and inventory for the Campus Closet platform
- Accelerated project delivery by developing a frontend for the Bridge OTW bed-allocation platform using TypeScript, Next.js, and Tailwind CSS, rapidly integrating Supabase backends via Jira-tracked workflows

Projects

Pointercrate Leaderboard | *C++, Geode SDK, Cocos2d-x* Jul. 2026 – Aug. 2026

- Created a custom C++ mod for the Cocos2d-x game engine, **integrating community-maintained leaderboard data** into Geometry Dash's native UI
- Manipulated pre-compiled game binaries via function hooking to safely **inject custom application logic**, managing dynamic memory allocation within legacy object structures **without source code access**

Raft Consensus Algorithm | *Golang* Oct. 2025 – Nov. 2025

- **Architected a fault-tolerant distributed system** in Golang based on the Raft Consensus Protocol to ensure **highly reliable state machine replication** across distributed nodes
- Implemented core distributed **networking protocols** including leader election, heartbeat mechanisms (via RPC), and log backtracking, guaranteeing strong consistency under simulated network partitions

MyFitnessTerrier | *Python, Next.js, TypeScript, MongoDB* Apr. 2025 – Sep. 2025

- **Engineered a full-stack web application** using TypeScript and Next.js, **enabling users to filter 600+ menu items** by 23 distinct nutrient parameters, dietary tags, and allergens
- **Aggregated daily nutrition data** across 5 university dining halls into MongoDB to **create a centralized campus dietary hub**
- **Reduced data-fetching latency from 120s to 30s** by restructuring the automated API extraction pipelines using Python and GitHub Actions